

Team No. 14

NEXUS HACKATHON

AI-POWERED COLLEGE KNOWLEDGE & CAMPUS ASSISTANT

A Smart Student Support System using RAG, AI & Campus Service

PROBLEM STATEMENT:

1. Scattered Information

- Notices
- Academic documents
- Timetables
- College information
- PDFs/web pages

2. Difficult to Find

- Students search across multiple sources
- Information takes time to locate
- Same question may require multiple sources.

3. Information Gap

- Unofficial information can cause confusion
- Time-sensitive updates can be missed.

THE PROBLEM IS NOT THE LACK OF INFORMATION - IT IS THE DIFFICULTY OF ACCESSING THE RIGHT INFORMATION AT THE RIGHT TIME.

OUR UNDERSTANDING:

- The challenge is **information accessibility**, not the lack of information.
- Students need a **single, trusted access point** for academic, administrative, and campus-related queries.

Our Objective:

- An Interface that should provide **official, contextual, and timely** information while reducing manual searching and improving the student experience.

MARKET RESEARCH:

College-Focused Platforms

Examples: CampusCopilot, CampusBot, Ocelot

- Focus on admissions or the broader student lifecycle.
- College-specific solutions exist.
- Their primary focus differs from our proposed campus-assistance workflow.

General Knowledge AI Platforms

Examples: Knowbase, Citai, Denser, Chat.co

- Strong document/knowledge-base capabilities.
- Support multiple data formats.
- Primarily positioned as general-purpose knowledge platforms.

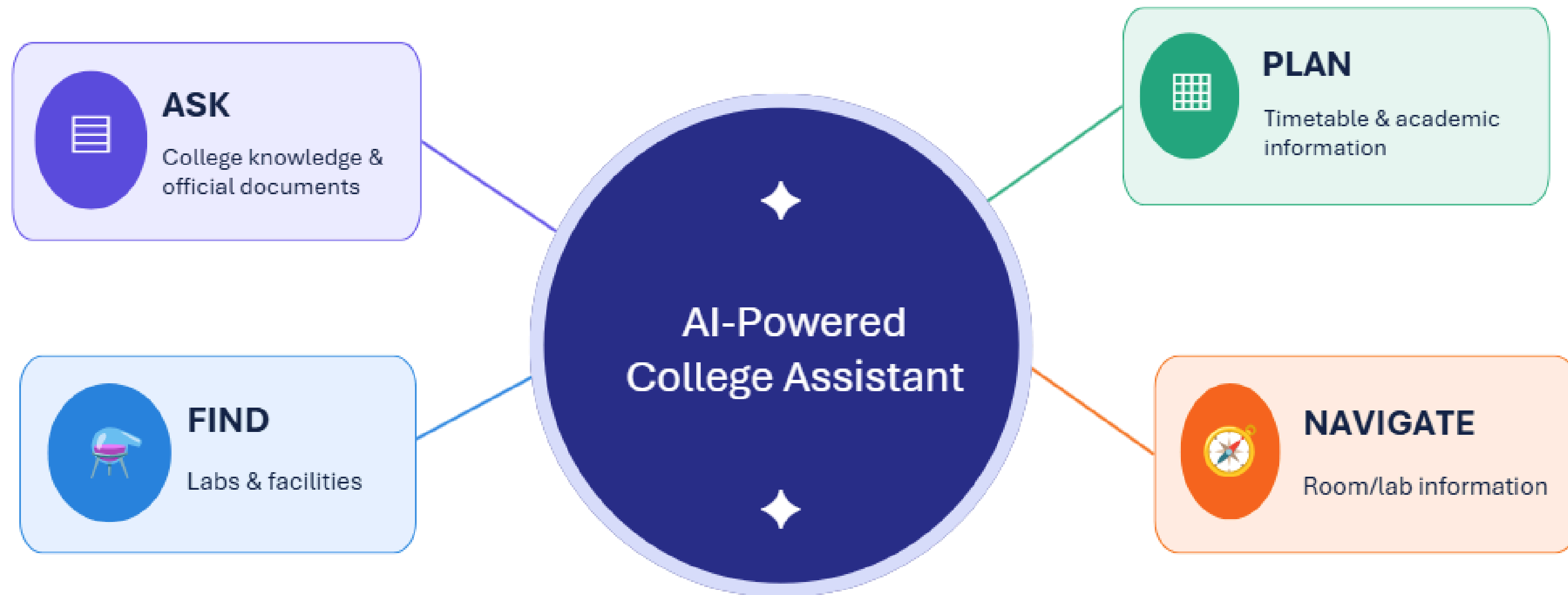
IDENTIFIED GAP:

Capability	Existing solution categories	Our proposed assistant
College knowledge	Available	Integrated
Document-based Q&A	Available	Integrated
Timetable assistance	Limited / varies	Integrated
Scheduled lab availability	Limited / varies	Integrated
Campus location/floor information	Limited / varies	Integrated
Unified conversational access	Limited / varies	Integrated

Most existing solutions focus on admissions, communication, or document-based knowledge. Our approach combines college knowledge with day-to-day campus assistance.

PROPOSED SOLUTION:

A Unified AI assistant that provides students with **official college information, timetable support and campus navigation** through a single intelligent platform.



One conversational interface for multiple campus needs.

USE CASE:

Example 1 - Knowledge

"What facilities does GLS University provide?"

- College documents / RAG

Example 2 - Location

"Which floor is room 507 on?"

- Campus location data

Example 3 - Labs

"Which labs are scheduled to be free at 2:15 PM?"

- Timetable + lab data

Example 4 - Academic information

"When does the semester start?"

- Official academic information

Example 5 - Timetable

"What classes do I have on Monday?"

- Timetable database

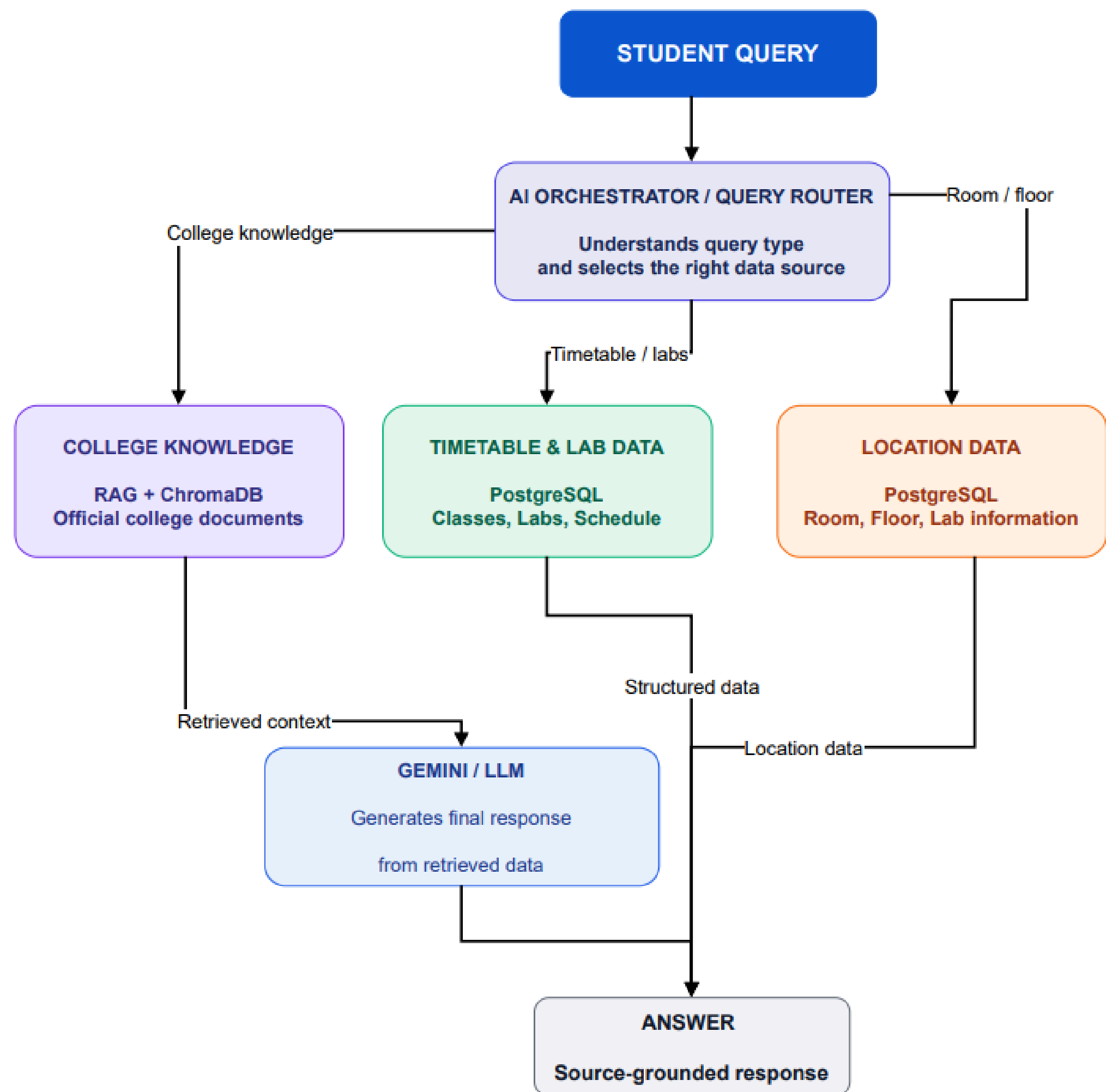
WHAT MAKES OUR SOLUTION DIFFERENT?

FROM SEARCHING FOR INFORMATION
TO GETTING THE RIGHT ANSWER

- **CONTEXT-AWARE** - Understands whether the student needs knowledge, schedule, lab or location information.
- **SOURCE-GROUNDED** - Uses official college information as the basis for responses.
- **ONE INTERFACE** - One interface for multiple campus needs.
- **LESS SEARCHING** - Reduces the need to search through multiple PDFs, pages and systems.

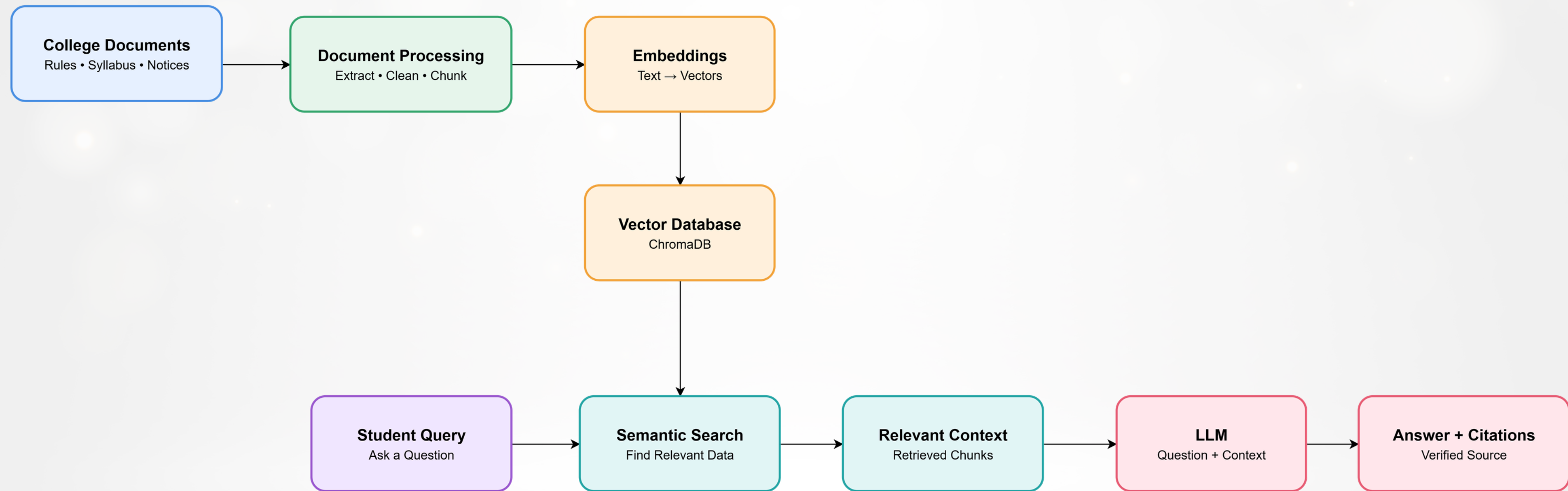
The goal is not just to answer a question - but to reduce the effort required to solve a student's campus-related problem.

WORKFLOW DIAGRAM:



RAG

WORKFLOW :



TOOLS & TECHNOLOGIES:

Frontend

- React.js
- CSS

Backend

- Node.js
- Express.js
- Python — RAG Service

AI & RAG

- Gemini
- ChromaDB - Vector Database
- Sentence Transformers

Database

- PostgreSQL - Structured Data

Development Tools

- Git
- GitHub
- Postman
- VS Code

Campus & Navigation

- Room & Location Lookup
- Campus/Floor Information



FUTURE SCOPE:

1. Smart Document Update Notifications

- Notify students when a document related to their previous query is updated or replaced.

2. Official Campus Floor Plan

- Integrate the college's official floor plan/campus map for easier room and building navigation.

3. Integration with More College Services

- Connect additional official college services and student resources through the same platform.



THANK YOU

Team No. 14

Resham Patel

Khushi Soneji